

Sjögren's Syndrome

A Complete Plain-English Guide

Comprehensive Reference for Patients, Caregivers, and Advocates

April 2026

Medical Disclaimer

This guide is for **educational purposes only** and does not constitute medical advice, diagnosis, or treatment. The information presented here is intended to help you understand Sjögren's Syndrome and have more informed conversations with your healthcare team. **Always consult a qualified healthcare professional** — such as a rheumatologist, ophthalmologist, or your primary care physician — before making any decisions about diagnosis, treatment, or medication changes. Every patient's situation is different, and only a clinician who knows your medical history can provide personalized guidance.

1. What Sjögren's Syndrome Actually Is

Sjögren's Syndrome (pronounced "SHOW-grins") is a **chronic autoimmune disease**. "Autoimmune" means your immune system — the defense network that is supposed to attack bacteria, viruses, and other invaders — turns against your own body instead.

In Sjögren's, the immune system's white blood cells — specifically **CD4+ T cells** (the "commander" cells that direct immune attacks) and **B cells** (the cells that produce antibodies) — invade and progressively destroy your **exocrine glands** (the glands that produce moisture). The main targets are the salivary glands (which make saliva) and the lacrimal glands (which make tears), but sweat glands, vaginal glands, and glands lining the respiratory and digestive tracts can also be attacked.

Primary vs. Secondary Sjögren's

- **Primary Sjögren's** — The disease stands alone. The patient does not have another autoimmune condition driving it.
- **Secondary Sjögren's** — The disease co-exists with another autoimmune disease, most commonly **rheumatoid arthritis**, **systemic lupus erythematosus (lupus)**, or **scleroderma**.

Who Gets It

Sjögren's predominantly affects **middle-aged women**, with the average age of diagnosis falling between 40 and 60 years old. However, men, children, and younger adults can develop it too. The estimated prevalence is roughly **1 in 1,600 people**, and the female-to-male ratio is approximately **9 to 1**.

Key Point

Sjögren's is best known for causing dryness — dry eyes and dry mouth. But it is a **systemic disease**, meaning it can attack organs and tissues throughout the entire body, including the joints, lungs, kidneys, nerves, blood vessels, and blood cells. Treating Sjögren's as "just a dryness disease" misses its full scope and can delay diagnosis of serious complications.

2. How It Damages Each Tissue Type

The immune system uses different mechanisms to attack different tissues in Sjögren's. Here is what happens in each major tissue category:

Glands (Salivary, Tear, Sweat, Vaginal)

Lymphocytes (a type of white blood cell) infiltrate the gland tissue in clusters. They directly destroy the **acinar cells** — the cells that actually produce moisture. Over time, the ducts (tiny tubes that carry moisture out of the gland) narrow and

scar. The gland gradually shrinks — a process called **atrophy**. The end result: the gland produces less and less moisture until output is severely reduced or stopped entirely.

Joints

Inflammatory chemicals called **cytokines** (signaling molecules released by immune cells) cause inflammation of the **synovium** (the thin lining inside joints). This is called **synovitis**. It causes pain, swelling, and stiffness — very similar to rheumatoid arthritis. The key difference: Sjögren's joint inflammation typically does **not** cause permanent bone erosion, whereas rheumatoid arthritis often does.

Skin

Immune complexes (clumps of antibodies stuck to their targets) deposit in the small blood vessels of the skin, triggering inflammation. This can cause rashes, purplish spots, and sometimes open ulcers. Separately, destruction of sweat glands leads to dry skin.

Lungs

Lymphocytes infiltrate lung tissue in two main patterns. **Interstitial lung disease** means inflammation and scarring develop in the tissue between the air sacs, reducing the lungs' ability to transfer oxygen. **Small-airway disease** means inflammation affects the tiny breathing tubes (bronchioles), causing cough and breathlessness.

Kidneys

Two distinct attack mechanisms occur. In **tubulointerstitial nephritis**, lymphocytes invade the tissue surrounding the kidney's tubes (tubules), disrupting the kidney's ability to balance electrolytes and acid. In **glomerulonephritis**, immune complexes deposit in the kidney's filtering units (glomeruli), directly impairing filtration. Both reduce kidney function, but through different pathways.

Nerves

Inflammation damages nerve fibers in multiple ways. **Large-fiber damage** causes weakness and numbness. **Small-fiber damage** causes burning pain, tingling, and prickling. The immune system can also directly attack the **nerve-cell bodies** in

clusters next to the spinal cord (called **dorsal root ganglia**), causing a condition known as **ganglionopathy** — loss of position sense and coordination.

Blood Vessels

Immune complexes and **cryoglobulins** (abnormal proteins that clump together when blood cools) deposit in vessel walls, causing inflammation called **vasculitis**. Vasculitis restricts blood flow, which can damage the skin, nerves, kidneys, and other organs downstream of the inflamed vessels.

Blood Cells and Immune Markers

Overactive B cells produce excessive antibodies — a condition called **hypergammaglobulinemia** (abnormally high levels of immune proteins in the blood). Over time, chronically stimulated B cells can undergo malignant transformation, evolving toward **lymphoma** (cancer of the lymphatic system). The immune system may also mistakenly attack the body's own blood cells, causing **leukopenia** (low white blood cell count) or **anemia** (low red blood cell count).

3. Symptom Explanations — What Each Symptom Actually Means

Below is a plain-English explanation of each major symptom cluster and the mechanism behind it.

Dry Eyes (Keratoconjunctivitis Sicca)

The tear glands stop producing enough tears. Without adequate tear film, the **cornea** (the clear front surface of the eye) and **conjunctiva** (the membrane lining the eyelids) dry out. This produces a gritty or sandy feeling, burning, sensitivity to light, and blurred vision that may fluctuate throughout the day.

Dry Mouth (Xerostomia)

The salivary glands are destroyed, so saliva production drops dramatically. Saliva does far more than keep your mouth wet — it protects teeth, fights bacteria, and aids digestion. Without it, patients experience difficulty swallowing, rapid tooth

decay, frequent mouth infections (especially **thrush**, a fungal infection), cracked lips, and altered or diminished taste.

Fatigue

This is chronic, profound exhaustion — not the same as normal tiredness or the fatigue you feel after a long day. It is caused by ongoing immune activation: the immune system is constantly "on," flooding the bloodstream with **inflammatory cytokines** (chemical signals that also trigger sickness behavior in the brain). Many patients describe it as the single most debilitating symptom.

Joint Pain and Stiffness

Inflammatory chemicals attack the synovial lining of joints, especially in the hands, wrists, and ankles. Morning stiffness — joints feeling locked up when you first wake — is common and typically improves with movement over the course of 30 to 60 minutes.

Skin Problems

Dry skin results from damaged sweat glands. Separately, blood vessel inflammation can produce **purpura** — purplish spots, typically on the lower legs — and various rashes. Some patients develop skin ulcers, particularly around the ankles.

Swollen Parotid Glands

The **parotid glands** — the large salivary glands located just in front of and below each ear — swell due to lymphocyte infiltration. The swelling often fluctuates, coming and going over weeks or months. **Persistently hard swelling that does not fluctuate is a warning sign for lymphoma and requires prompt evaluation.**

Vaginal Dryness

The same gland-destruction mechanism that attacks salivary and tear glands also targets the moisture-producing glands of the vaginal lining. This causes dryness, irritation, and pain during intercourse.

Difficulty Swallowing (Dysphagia)

The glands that keep the esophagus (the tube from mouth to stomach) moist are destroyed, and the mucosal lining atrophies (thins out). Food feels like it sticks in the throat, especially dry or crumbly foods.

Cough and Shortness of Breath

Airway dryness — from destruction of glands lining the respiratory tract — causes a dry, persistent cough. If interstitial lung disease develops, patients also experience progressive shortness of breath with exertion.

Brain Fog and Cognitive Difficulty

Many patients report difficulty concentrating, memory lapses, and mental sluggishness. The mechanism is not fully understood, but it is linked to inflammatory cytokines crossing into or affecting the central nervous system.

Burning, Tingling, or Numbness in Hands and Feet

This is **peripheral neuropathy** — damage to nerve fibers outside the brain and spinal cord. The burning and tingling are caused by small-fiber nerve damage. Numbness and weakness indicate large-fiber involvement.

Raynaud's Phenomenon

Blood vessels in the fingers and toes spasm excessively in response to cold or stress. The affected digits turn white (blood supply cuts off), then blue (deoxygenated blood pools), then red (blood rushes back). It can be painful and, in severe cases, cause small sores on the fingertips.

4. All Treatment Categories

Treatments are organized from mildest to strongest. Most patients start with moisture replacement and add systemic medications only if the disease involves organs beyond the glands.

4A. Moisture Replacement (Symptomatic Relief)

These treatments do not slow the disease itself — they replace what the damaged glands can no longer produce.

- **Artificial tears** — Over-the-counter eye drops. Preservative-free formulations are recommended for patients who use them more than four times per day, because preservatives themselves can irritate the eye surface over time.
- **Prescription anti-inflammatory eye drops** — **Cyclosporine** (Restasis) and **lifitegrast** (Xiidra) reduce inflammation on the eye surface, improving the quality and quantity of natural tears. These take several weeks to reach full effect.
- **Punctal plugs** — Tiny silicone or collagen plugs inserted into the tear ducts (the small drainage holes in the inner corners of the eyelids). By blocking drainage, tears stay on the eye surface longer.
- **Saliva substitutes** — Sprays or gels that coat the mouth and mimic saliva.
- **Sugar-free gum and lozenges** — Stimulate whatever salivary gland tissue remains to produce more saliva.
- **Prescription saliva stimulants** — **Pilocarpine** (Salagen) and **cevimeline** (Evoxac) are oral medications that stimulate remaining gland tissue to produce more moisture across multiple gland types (salivary, tear, sweat). Common side effects include sweating and flushing.
- **Vaginal moisturizers and lubricants** — Over-the-counter products that restore vaginal moisture.
- **Skin moisturizers** — Emollient creams (e.g., Eucerin, Lubriderm) applied frequently, especially after bathing.

4B. Anti-Inflammatory and Disease-Modifying Medications

- **Hydroxychloroquine (Plaquenil)** — The first-line systemic drug for Sjögren's. It calms the overactive immune system broadly and is most effective for joint pain, fatigue, and skin rashes. It takes **3 to 6 months** to reach full effect. Requires regular eye exams (typically annually) to monitor for a rare side effect: retinal toxicity that can affect vision.
- **NSAIDs** (ibuprofen, naproxen) — Non-steroidal anti-inflammatory drugs. Used for joint pain and inflammation. Available over the counter, but long-term use carries risks to the stomach, kidneys, and cardiovascular system.
- **Low-dose corticosteroids (prednisone)** — Powerful anti-inflammatory drugs used for short-term control of flares. Not ideal for long-term use because of significant side effects (bone loss, weight gain, blood sugar elevation, immune suppression, cataracts).

4C. Immunosuppressants (for Organ Involvement)

These medications suppress the immune system more aggressively and are reserved for patients whose disease is attacking organs.

- **Methotrexate** — Used for inflammatory arthritis or lung involvement. Taken weekly. Requires monitoring of liver function and blood counts.
- **Mycophenolate mofetil (CellCept)** — Used for kidney or lung inflammation. Suppresses the proliferation of lymphocytes.
- **Azathioprine (Imuran)** — An alternative immunosuppressant for organ involvement. Requires blood count monitoring.
- **Cyclophosphamide** — The strongest conventional immunosuppressant. Reserved for severe, life-threatening organ involvement such as severe lung disease or severe vasculitis. Significant side effects, including infection risk and bladder toxicity.

4D. Biologic Therapies

- **Rituximab (Rituxan)** — A monoclonal antibody (a lab-made protein designed to target a specific molecule) that targets **CD20**, a marker on the surface of B cells. It depletes B cells from the bloodstream. Used for severe vasculitis, lymphoma, severe parotid swelling, peripheral neuropathy (especially **mononeuritis multiplex**), pulmonary disease, and symptoms that do not respond to other treatments. Given as an intravenous infusion.
- **Pipeline drugs in clinical trials** — Several new biologics are under investigation:
 - **Ianalumab** — targets the BAFF receptor (a survival signal for B cells)
 - **Iscalimab** — targets CD40 (a molecule involved in B-cell activation)
 - **Dazodalibep** — blocks CD40 ligand (the partner molecule for CD40)
 - **Remibrutinib** — a BTK inhibitor (blocks a signaling enzyme inside B cells)

4E. Supportive Care

- **Dental care** — Fluoride treatments (prescription-strength fluoride trays or rinses) and dental visits every 6 months or more frequently. Dry mouth dramatically accelerates tooth decay, so proactive dental care is essential.
- **Humidifiers** — For home and office to maintain ambient moisture, especially during sleep.
- **Sunglasses and moisture-chamber glasses** — Wraparound eyewear that reduces tear evaporation by shielding eyes from wind and dry air.
- **Physical therapy** — For joint stiffness and maintaining range of motion.

5. All Tests for Diagnosis and Organ Involvement

5A. Blood Tests for Diagnosis

Test	What It Measures	What It Means in Sjögren's
ANA (Antinuclear Antibody)	Antibodies that attack the cell nucleus	Positive in most patients. A speckled pattern is typical. But a positive ANA alone does not confirm Sjögren's — many healthy people and people with other conditions have a positive ANA. It is a screening tool, not a definitive test.
Anti-SSA (Ro) antibodies	Antibodies targeting the SSA/Ro protein	Found in 60–80% of patients. The single most important antibody for diagnosis. A positive SSA with compatible symptoms strongly supports the diagnosis.
Anti-SSB (La) antibodies	Antibodies targeting the SSB/La protein	Found in 30–50% of patients. Almost always appears alongside SSA — it is rare to have SSB without SSA. Having both SSA and SSB is associated with more severe disease.
Rheumatoid Factor (RF)	An antibody that attacks other antibodies	Positive in many patients. Also serves as a marker for lymphoma risk when persistently elevated.
ESR (Erythrocyte Sedimentation Rate)	How fast red blood cells settle in a tube — a general inflammation marker	Often elevated in active disease. Non-specific but useful for tracking disease activity over time.

Test	What It Measures	What It Means in Sjögren's
Immunoglobulin levels (IgG, IgA, IgM)	Levels of the three main antibody types	Often elevated (hypergammaglobulinemia) due to overactive B cells producing excess antibodies.
Complete Blood Count (CBC)	Counts of red cells, white cells, and platelets	Checks for anemia (low red cells), leukopenia (low white cells), and low platelets — all of which can result from immune system attack on blood cells.
Complement levels (C3, C4)	Levels of complement proteins (part of the immune defense system)	Low C4 is a warning sign — it is associated with vasculitis and increased lymphoma risk. Complement gets "used up" when immune complexes are forming and depositing in tissues.
Cryoglobulins	Abnormal proteins that clump together when blood is cooled	Present in cryoglobulinemic vasculitis. A strong lymphoma risk marker . Requires a special blood draw (kept warm during transport to the lab).
Beta-2 microglobulin	A protein shed from cell surfaces, elevated when lymphocytes are highly active	Elevated levels are associated with increased lymphoma risk and higher disease activity.

5B. Eye Tests

- **Schirmer test** — A small strip of filter paper is placed on the inside of the lower eyelid for 5 minutes. The strip absorbs tears, and the length of wetting is measured. Less than **5 mm of wetting** is considered abnormal.
- **Ocular staining score** — Dyes (**fluorescein** and **lissamine green**) are applied to the eye surface. Damaged areas of the cornea and conjunctiva

absorb the dye and become visible under a slit lamp. The extent of staining is scored to measure the severity of eye-surface damage.

- **Tear break-up time** — After dye is applied, the doctor watches through a slit lamp to see how quickly the tear film breaks apart. A short break-up time means the tears are unstable and evaporate too fast.

5C. Mouth and Salivary Tests

- **Unstimulated salivary flow rate** — The patient spits into a collection tube for 15 minutes without any stimulation. Less than **1.5 mL** of saliva collected is considered abnormal.
- **Salivary gland ultrasound** — A non-invasive imaging technique that shows characteristic patterns of gland damage — scattered dark spots (areas where gland tissue has been replaced by lymphocyte clusters) within the gland. Increasingly used as a first-line diagnostic tool.
- **Minor salivary gland (lip) biopsy** — A small tissue sample is taken from the inside of the lower lip under local anesthesia. A pathologist examines the tissue under a microscope and counts clusters of lymphocytes. The result is reported as a **focus score**: one "focus" is a cluster of 50 or more lymphocytes. A focus score of **1 or more per 4 square millimeters** confirms the diagnosis. This is often considered the gold standard when antibodies are negative.

5D. Organ-Involvement Tests

Kidney

- **BUN and creatinine** — Blood tests that measure kidney filtration function.
- **Urinalysis** — Looking for protein, blood, or abnormal pH in the urine.
- **Urine pH** — Persistently alkaline urine (pH above 5.5 when the blood is acidic) suggests **renal tubular acidosis** (the kidneys are unable to properly excrete acid).

- **Electrolytes** — Low potassium is a clue to kidney tubule damage.
- **Kidney biopsy** — If needed to determine the type and severity of kidney inflammation and guide treatment decisions.

Lungs

- **Chest X-ray** — Initial screening for lung abnormalities.
- **High-resolution CT scan** — Detailed imaging that can detect interstitial lung disease and small-airway disease.
- **Pulmonary function tests** — Breathing tests that measure lung capacity and airflow. Looking for restrictive patterns (lungs can't expand fully, suggesting scarring) or obstructive patterns (air can't flow out properly, suggesting airway disease).

Nerves

- **Nerve conduction study / EMG (electromyography)** — Electrical tests that assess large-fiber nerve function. These tests are **normal** in pure small-fiber neuropathy, which is the most common nerve problem in Sjögren's.
- **Skin punch biopsy for small-fiber neuropathy** — A tiny skin sample (usually from the lower leg) is examined under a microscope to count the density of nerve endings. Reduced nerve-fiber density confirms small-fiber neuropathy.
- **Tilt-table test** — The patient lies on a table that is tilted upright while heart rate and blood pressure are monitored. Used to diagnose autonomic dysfunction (e.g., POTS — postural orthostatic tachycardia syndrome).

Blood Vessels

- **Skin biopsy** — A biopsy of purpura or rash to confirm vasculitis and determine the type of inflammation.
- **Cryoglobulin testing** — Specialized blood test (see Section 5A above).

Lymphoma Surveillance

- **Physical exam** — Palpating (feeling) lymph nodes and parotid glands for enlargement or hardening.
- **SPEP (Serum Protein Electrophoresis) and immunofixation** — Blood tests that separate proteins by type. Detect abnormal protein bands that may indicate a lymphoma or pre-lymphoma condition.
- **Free light chains** — Measures fragments of antibodies in the blood. Abnormal ratios suggest B-cell disorders.
- **Imaging (CT, PET)** — Used if suspicious swelling is found to determine the extent and location of any abnormal tissue.
- **Biopsy** — Of any persistently enlarged gland or lymph node. This is the only definitive way to diagnose or rule out lymphoma.

6. Severe-Case Priorities — What Matters Most When It Gets Serious

When Sjögren's moves beyond dryness symptoms, the following complications are listed in order of clinical urgency — the ones that demand the fastest action are at the top.

Priority	Complication	Why It Is Urgent	Action Required
1	Lymphoma development	B-cell lymphoma (usually MALT type) is the most serious long-term risk of Sjögren's. A persistently hard or growing parotid gland or lymph node that stops fluctuating is the cardinal warning	Urgent biopsy. Oncology/hematology referral. Imaging for staging.

Priority	Complication	Why It Is Urgent	Action Required
		sign.	
2	Vasculitis with organ damage	Cryoglobulinemic vasculitis can damage kidneys, nerves, and skin simultaneously. Without treatment, organ damage can become permanent.	Rituximab. Cyclophosphamide in life-threatening cases. Urgent nephrology and neurology referrals.
3	Rapidly worsening kidney function	Acute interstitial nephritis or aggressive glomerulonephritis can cause permanent kidney damage if not treated quickly.	Immediate immunosuppression. Kidney biopsy to guide treatment. Nephrology referral.
4	Severe neuropathy	Mononeuritis multiplex (sudden nerve damage in multiple locations) or ganglionopathy (loss of position sense and coordination) can cause severe, irreversible disability.	Urgent neurology referral. Aggressive immunosuppression (rituximab, IVIG, corticosteroids).
5	Interstitial lung disease	Progressive scarring of lung tissue reduces breathing capacity. The damage is often irreversible once scarring is established.	Pulmonary function monitoring. Immunosuppression (mycophenolate, rituximab). Pulmonology referral.
6	Severe systemic flare	Fever, weight loss, and night sweats combined with blood abnormalities may signal either severe disease activity or	Comprehensive workup to distinguish flare from lymphoma. Consider hospitalization. Urgent blood work, imaging,

Priority	Complication	Why It Is Urgent	Action Required
		transformation to lymphoma.	and biopsy if indicated.

7. Kidney Involvement — In Detail

Kidney disease occurs in about **5% of Sjögren's patients**. While that makes it relatively uncommon, it is one of the more consequential complications because kidney damage can be silent until it is advanced.

Tubulointerstitial Nephritis (TIN)

This is the **most common form of kidney involvement** in Sjögren's.

Lymphocytes invade the tissue surrounding the kidney's tubules — the tiny tubes that reabsorb water, electrolytes, and nutrients after the blood has been filtered. When these tubules are damaged, they lose the ability to reabsorb electrolytes properly and to maintain the body's acid-base balance. The result is often **renal tubular acidosis** (described below) and **hypokalemia** (low potassium). Symptoms can include muscle weakness, fatigue, and in severe cases, **paralysis** from dangerously low potassium levels.

Renal Tubular Acidosis (RTA)

The damaged kidney tubules fail to excrete acid properly, so acid builds up in the blood. This is typically **Type 1 (distal) RTA**. It may be present before any obvious kidney inflammation is visible on biopsy — meaning it can be an early sign. RTA can also cause **kidney stones** because the alkaline urine favors stone formation.

Glomerulonephritis

Less common than TIN. Immune complexes deposit in the kidney's glomeruli (filtering units), causing inflammation that damages the filters themselves. This can cause **protein in the urine** (proteinuria), **blood in the urine** (hematuria), and progressive kidney damage. Glomerulonephritis in Sjögren's is often associated with **cryoglobulinemia**.

Monitoring

Regular blood tests — creatinine (kidney filtration marker), electrolytes (especially potassium), and bicarbonate (acid-base balance) — plus urinalysis and urine pH. Early detection prevents chronic, irreversible damage.

Treatment

Electrolyte correction is the immediate priority: **potassium supplementation** and **bicarbonate supplementation** (to neutralize excess acid). **Corticosteroids** are used to reduce the immune attack on the kidneys. For more severe or refractory cases, stronger immunosuppression with **mycophenolate** or **rituximab** may be required. A **kidney biopsy** guides treatment decisions by revealing the type and severity of inflammation.

8. Nerve Involvement — In Detail

Nerve involvement affects **10-20% of Sjögren's patients** by conventional estimates, but some studies suggest the number may be as high as **60%** when small-fiber neuropathy is carefully tested for. Critically, nerve symptoms can appear **before dryness symptoms** in up to 90% of cases — meaning some patients see neurologists long before anyone suspects Sjögren's.

Small-Fiber Neuropathy (SFN)

The **most common nerve problem** in Sjögren's. The tiny nerve endings in the skin are damaged, causing burning, stabbing pain, prickling sensations, and abnormal sensitivity to touch — typically in the feet and hands. Standard nerve-conduction tests come back **completely normal** because those tests only measure large fibers. Diagnosis requires a **skin punch biopsy** that counts nerve-fiber density under a microscope. Because of the normal nerve-conduction results, SFN is frequently **missed or dismissed** as anxiety or "all in your head."

Large-Fiber Neuropathy

Damages the bigger, myelinated nerve fibers. Causes numbness, weakness, and loss of reflexes. Unlike small-fiber neuropathy, this **is** detected by standard nerve-conduction studies and EMG.

Sensory Ganglionopathy

The immune system attacks the **nerve-cell bodies** in the dorsal root ganglia — clusters of nerve cells located next to the spinal cord. This causes loss of **proprioception** (position sense — the ability to know where your limbs are without looking). Patients lose coordination, have difficulty walking, and may develop unsteady, wide-based gaits. This can be severely disabling.

Autonomic Neuropathy

Damages the nerves controlling involuntary functions — heart rate, blood pressure, sweating, digestion, and bladder control. Manifestations include:

- **POTS** (postural orthostatic tachycardia syndrome) — rapid heart rate upon standing
- Low blood pressure upon standing (orthostatic hypotension)
- Abnormal sweating patterns
- Digestive problems (gastroparesis, constipation)
- Bladder dysfunction

Autonomic neuropathy is **often misdiagnosed as anxiety** because symptoms like racing heart, dizziness, and digestive upset overlap with anxiety presentations.

Cranial Neuropathies

Inflammation of specific cranial nerves. The most commonly affected are the **trigeminal nerve** (causing facial numbness or pain) and the **facial nerve** (causing facial weakness or drooping on one side).

Mononeuritis Multiplex

Vasculitis-driven damage to individual nerves in different areas of the body simultaneously — for example, sudden wrist drop in one hand and foot drop on the opposite side. This is a **medical emergency** requiring aggressive immunosuppression because it indicates active vasculitis destroying nerves.

Treatment

Treatment depends on the type of neuropathy:

- **Pain medications** — gabapentin, pregabalin, or duloxetine for symptomatic relief of burning and tingling.
- **Immunosuppression** — corticosteroids, IVIG (intravenous immunoglobulin — pooled antibodies from donors that can modulate the immune response), or rituximab for inflammatory and vasculitic neuropathies.

9. Blood-Vessel Involvement (Vasculitis) — In Detail

Vasculitis — inflammation of blood vessel walls — occurs in about **10% of Sjögren's patients**. It is one of the disease's most serious complications because inflamed blood vessels can restrict blood flow to any organ they supply.

Cutaneous (Skin) Vasculitis

The most common form. Immune complexes deposit in small blood vessels in the skin, triggering inflammation. The signature finding is **palpable purpura** — small, raised, purplish spots that you can feel with your fingertips, most often appearing on the lower legs and ankles. In severe cases, skin ulcers develop.

Cryoglobulinemic Vasculitis

The **most dangerous form** of vasculitis in Sjögren's. Abnormal proteins called **cryoglobulins** (specifically **Type II mixed cryoglobulins**) circulate in the blood. When the blood cools — such as in the extremities — these proteins clump together and deposit in vessel walls, triggering inflammation. Cryoglobulinemic vasculitis typically causes a triad of problems simultaneously: purpura, kidney damage, and neuropathy. Patients with cryoglobulinemic vasculitis have a **6-fold increased risk of death or lymphoma** compared to Sjögren's patients without it.

Hypergammaglobulinemic Vasculitis

Caused by excess antibodies (immunoglobulins) depositing in vessel walls. It causes purpura similar to cryoglobulinemic vasculitis but is generally **less severe** and carries a lower risk of organ damage and lymphoma.

Systemic Vasculitis

Can affect blood vessels in multiple organs simultaneously — kidneys, lungs, nerves, and gastrointestinal tract. This is the broadest and most dangerous pattern.

Warning Signs of Vasculitis

- New purplish spots on the legs (especially if you can feel them as raised bumps)
- Skin ulcers, particularly around the ankles
- Worsening neuropathy symptoms (new numbness, weakness, or burning pain)
- New kidney problems (protein in urine, rising creatinine, low potassium)

Treatment

- **Mild cutaneous vasculitis** — Corticosteroids.
- **Cryoglobulinemic vasculitis** — **Rituximab** is the primary treatment.
- **Life-threatening vasculitis** — **Cyclophosphamide** may be required.
- **Hepatitis C screening** — Hepatitis C infection can independently cause cryoglobulinemia. All patients with cryoglobulins should be tested for hepatitis C, because the treatment approach differs if the virus is present.

10. Lymphoma-Risk Monitoring

Sjögren's patients have a **5-10% lifetime risk** of developing **non-Hodgkin lymphoma** — roughly **15 to 44 times higher** than the general population. This is the single most serious long-term risk of the disease and the primary reason for ongoing monitoring.

Types of Lymphoma in Sjögren's

- **MALT lymphoma** (Mucosa-Associated Lymphoid Tissue) — The most common type. It is typically slow-growing and usually arises in the salivary glands. It develops from the same B cells that have been chronically stimulated by the autoimmune process.
- **Diffuse Large B-Cell Lymphoma (DLBCL)** — Less common but more aggressive. May arise on its own or as a transformation from a pre-existing MALT lymphoma.

Risk Factors That Increase Lymphoma Probability

Risk Factor	Why It Matters
Persistently low C4 complement levels	Indicates ongoing immune-complex formation and chronic B-cell activation
Presence of cryoglobulins	Marker of abnormal B-cell behavior; strong predictor of lymphoma
Persistent parotid gland swelling	Indicates heavy lymphocyte infiltration of the gland — the tissue where MALT lymphoma most often arises
Palpable purpura (vasculitis)	Reflects immune-complex-driven inflammation associated with B-cell dysregulation
High rheumatoid factor	Another marker of chronic B-cell stimulation
Lymph node enlargement	May represent reactive enlargement or early lymphoma
Low white blood cell count (lymphopenia)	Associated with more severe autoimmune activity and higher lymphoma risk
Elevated beta-2 microglobulin	Marker of high lymphocyte turnover — elevated in both active disease and lymphoma

Clinical Warning Signs Requiring Immediate Evaluation

Seek Immediate Medical Evaluation If You Notice:

- A parotid gland that becomes persistently hard, keeps growing, or stops fluctuating in size
- New or enlarging lymph nodes that do not go away after several weeks
- Unexplained weight loss (more than 10% of body weight)
- Drenching night sweats (soaking through clothes or bedsheets)
- Persistent unexplained fever
- Rapidly worsening fatigue beyond your baseline

Monitoring Schedule

Regular physical exams with palpation of parotid glands and lymph nodes. Blood work every **6 to 12 months** including CBC, complement levels (C3, C4), cryoglobulins, immunoglobulin levels, and SPEP. Any suspicious tissue — a hard gland, a growing lymph node, a persistent mass — warrants **prompt biopsy**. Early detection of lymphoma dramatically improves treatment outcomes.

11. Severity-Based Doctor Checklist

Use the following checklist as a framework for discussions with your healthcare team. The appropriate tier depends on your current disease activity and organ involvement.

Tier 1 — Routine Monitoring

Applicable when: Dryness symptoms only, no organ involvement.

Frequency: See rheumatologist every 6–12 months.

Category	Action Items
Blood work	CBC, metabolic panel, ESR, immunoglobulins (IgG, IgA, IgM), C3/C4 complement, ANA/SSA/SSB
Eye care	Annual ophthalmologist exam including Schirmer test and ocular staining score
Dental care	Dental exam every 6 months with fluoride treatments
Kidney screening	Urinalysis and electrolytes to screen for early kidney involvement

Tier 2 — Active Disease Management

Applicable when: Joint inflammation, significant fatigue, skin involvement, mild neuropathy symptoms.

Frequency: See rheumatologist every 3–6 months.

Category	Action Items
Blood work	All Tier 1 tests plus rheumatoid factor, cryoglobulins
Medication review	Consider hydroxychloroquine if not already started
Nerve screening	Ask about burning, numbness, dizziness on standing — screen for neuropathy at every visit
Lung screening	Pulmonary function tests if any cough or breathlessness is reported
Gland imaging	Salivary gland ultrasound if swelling is present
Eye and dental care	Continue all Tier 1 eye and dental monitoring

Tier 3 — Organ Involvement

Applicable when: Kidney abnormalities, confirmed neuropathy, lung involvement, vasculitis.

Frequency: See rheumatologist every 1-3 months, plus relevant specialists.

Category	Action Items
Blood work	All Tier 1 and 2 tests with increased frequency
Nephrology	Kidney biopsy consideration, renal tubular acidosis workup, regular electrolyte monitoring (potassium, bicarbonate)
Neurology	Nerve conduction studies, skin biopsy for small-fiber neuropathy, tilt-table test for autonomic dysfunction
Pulmonology	High-resolution CT scan, pulmonary function tests, oxygen saturation monitoring
Dermatology	Skin biopsy of any purpura or rash to confirm vasculitis
Treatment escalation	Immunosuppressive therapy: methotrexate, mycophenolate, or azathioprine as appropriate for the specific organ involvement

Tier 4 — Severe / Emergency

Applicable when: Suspected lymphoma, cryoglobulinemic vasculitis, rapidly declining kidney function, mononeuritis multiplex, severe interstitial lung disease.

Frequency: Immediate specialist attention.

Category	Action Items
Biopsy	Urgent biopsy of any persistently hard or growing gland or lymph node
Oncology / Hematology	Referral if lymphoma is suspected based on clinical findings or blood work

Category	Action Items
Aggressive treatment	Rituximab or cyclophosphamide consideration based on the specific emergency
Hospitalization	If acute organ failure is occurring or imminent
Advanced imaging	CT and/or PET scan for staging of suspected lymphoma
Advanced blood work	SPEP, immunofixation, free light chains, beta-2 microglobulin

Final Note

This guide is for **educational purposes only**. It is designed to help you understand the mechanisms, symptoms, tests, and treatments associated with Sjögren's Syndrome so that you can be an informed participant in your own healthcare. It is not a substitute for professional medical advice, diagnosis, or treatment. All medical decisions — including starting, stopping, or changing medications — should be made in partnership with a qualified healthcare professional who knows your individual history and circumstances.

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